Robust Crack Detection with Few Labels

Abstract. We study crack detection on concrete surfaces when only a small share of images is labelled. A segmentation network is pre-trained with self-supervision and then fine-tuned on the labelled subset.

1 Introduction. Visual inspection of bridges is slow and expensive. Labelled data is scarce because annotating thin cracks pixel by pixel takes experts several minutes per image.

2 Method. The encoder is a ResNet-34. Fine-tuning uses the AdamW optimiser with a learning rate of 3e-4, a batch size of 32 and 40 epochs. Images are cropped to 512 by 512 pixels.

3 Dataset. We collected 12,400 images from 31 bridges. Of these, 9,300 are used for training and 3,100 for testing. Only 10 percent of the training images carry labels.

4 Results. Our model reaches an F1 score of 87.4 percent on the test set. The supervised baseline trained on the same labelled images reaches 81.2 percent. Without the augmentation step the F1 score of our model drops to 84.9 percent.

5 Limitations. The images come from daylight inspections only. Performance on night images and on wet surfaces was not evaluated.

6 Conclusion. Self-supervised pre-training closes most of the gap to fully supervised training while using a tenth of the labels.